

Radiation Effects
Space & High-Energy Particle Environments

Single-Event Effects
Ion or proton strike in sensitive volume

Cumulative Effects
Long-duration exposure degradation

SEU
Single-Event Upset

Bit-flip in memory cell or register; soft error (non-destructive, recoverable)

SET
Single-Event Transient

Spurious voltage pulse in analog or combinational logic; may propagate as a soft error

SEFI
SE Functional Interrupt

Upset in control logic (FSM, config register); disrupts operation; reset-recoverable

SEL
Single-Event Latch-up

Parasitic PNP (SCR) triggered; high current state; potentially destructive

SEB
Single-Event Burnout

Destructive thermal runaway in power MOSFETs & bipolar devices

SEGR
SE Gate Rupture

Gate oxide dielectric breakdown; permanent device failure in power MOSFETs

TID
Total Ionizing Dose

Charge trapping in oxide; Vth shift and leakage increase (krad / Mrad metric)

DD
Displacement Damage

Crystal lattice defects; minority carrier lifetime reduction (NIEL MeV·cm²/g metric)

Soft / Recoverable (SEU, SET, SEFI)

Potentially Destructive (SEL, SEB, SEGR)

Cumulative / Permanent Degradation (TID, DD)